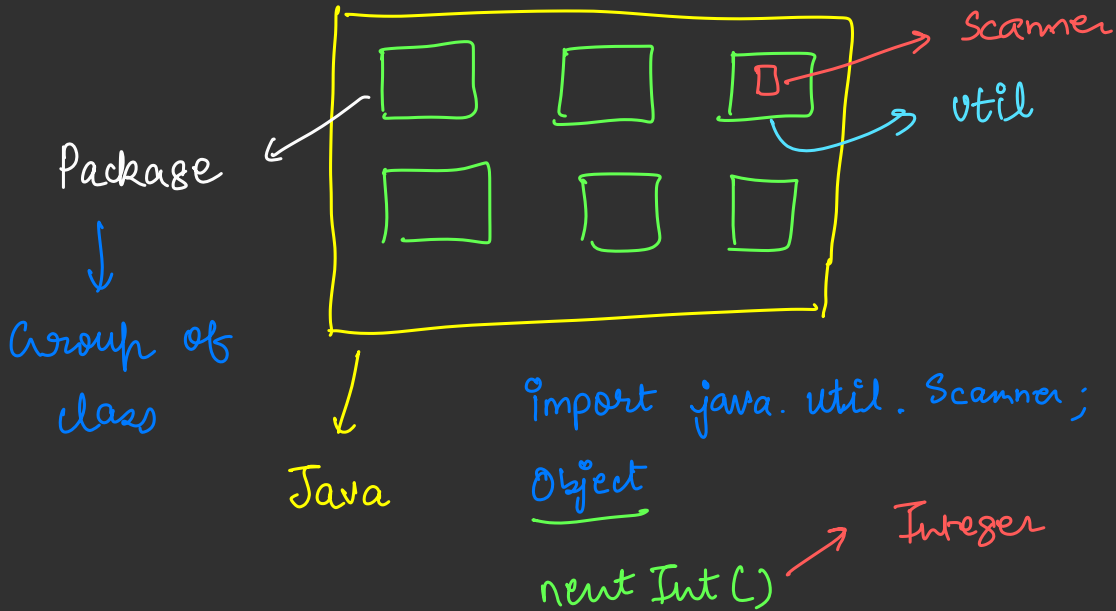


L3

Control flow (if-else & loops)

Join Discord - <https://tinyurl.com/ly-discord>



$$a = 5$$

$$b = 6$$

$$a + b$$

$$a - b$$

$$a * b$$

$$a / b$$

$$a \% b$$



$$5 \% 6 \rightarrow$$

$$\begin{array}{r} 6 \overline{) 50} \\ \underline{-0} \\ 5 \end{array}$$

Arithmetic Operator
+, -, *, /, %

$$\text{double } a = 5.0;$$

$$\text{double } b = 6.0;$$

$$\text{double } c = a / b;$$

$$c = \frac{5.0}{6.0} = \underline{\underline{0.83}}$$

$$\underline{\underline{5 \% 6 = 5}}$$

$8 \% 5$ $\rightarrow$ Remainder

$$\begin{array}{r} 5 \overline{) 8} 1 \\ - 5 \\ \hline 3 \end{array}$$

$$8 / 5 = 1$$

$$\begin{array}{r} 5 \overline{) 6} 1 \\ - 5 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 6 \overline{) 5} 0 \\ - 0 \\ \hline 5 \end{array}$$

int $\rightarrow -2^{*} 10^9$ to $2^{*} 10^9$

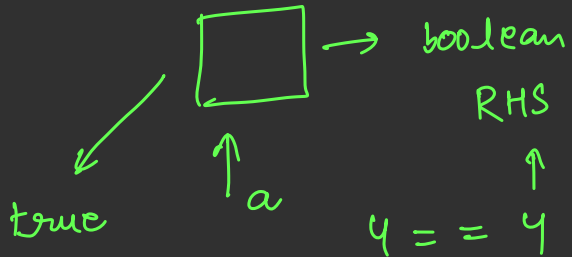
long $\rightarrow -2^{*} 10^{18}$ to $2^{*} 10^{18}$

double

boolean a = true;

float

boolean $\rightarrow$ true
boolean $\rightarrow$ false



5 == 6
 $\uparrow$ $\uparrow$
LHS RHS $\rightarrow$ false

4 == 4
 $\downarrow$ $\uparrow$
LHS true

true / false $\leftarrow$ Relational operator

$= =$

value1 == value2

LHS > RHS
true

$\rightarrow$ true
 $\rightarrow$ false

5 > 4



true

5 > 6



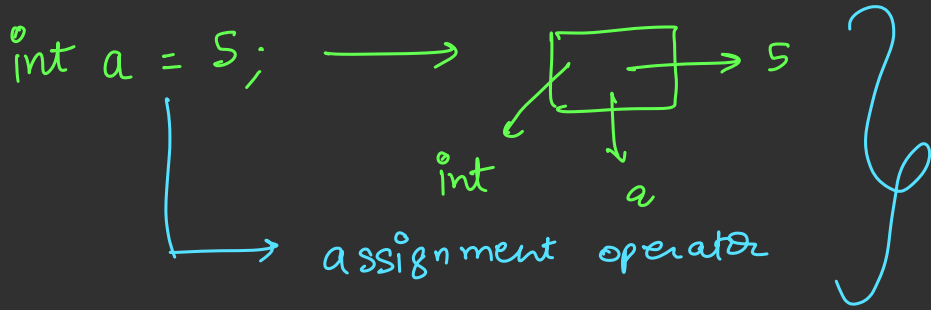
false

>

<

LHS < RHS $\rightarrow$ true

$\rightarrow$ false



$>$ $5 > 5$ $\longrightarrow$ false

$> =$ $5 > = 5$ $\longrightarrow$ true

$\uparrow$ LHS $\uparrow$ RHS

$6 > = 5$

$< =$

$4 < = 5 \rightarrow \text{true}$

$4 < = 4 \rightarrow \text{true}$

$6 < = 5 \rightarrow \text{false}$

$! =$

$\rightarrow \text{LHS} \neq \text{RHS} \rightarrow \text{true}$

$\swarrow$

$\hookrightarrow$

different

$\nearrow$

true

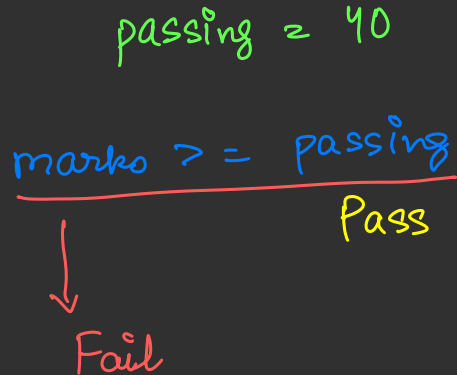
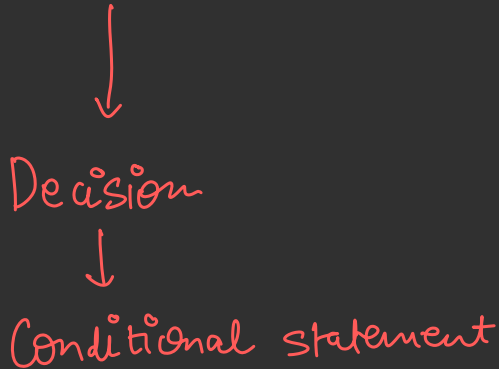
$6 \neq 4$

same = false

$4 \neq 4 \rightarrow \text{false}$

Let's suppose you want the computer to decide whether someone will pass or fail in exam based on marks entered.

How do you think we can do it ?



What are conditionals statements and their use ?

→ if

→ if else

→ if else if

→ switch case

→ Decision making

If → Keyword

value

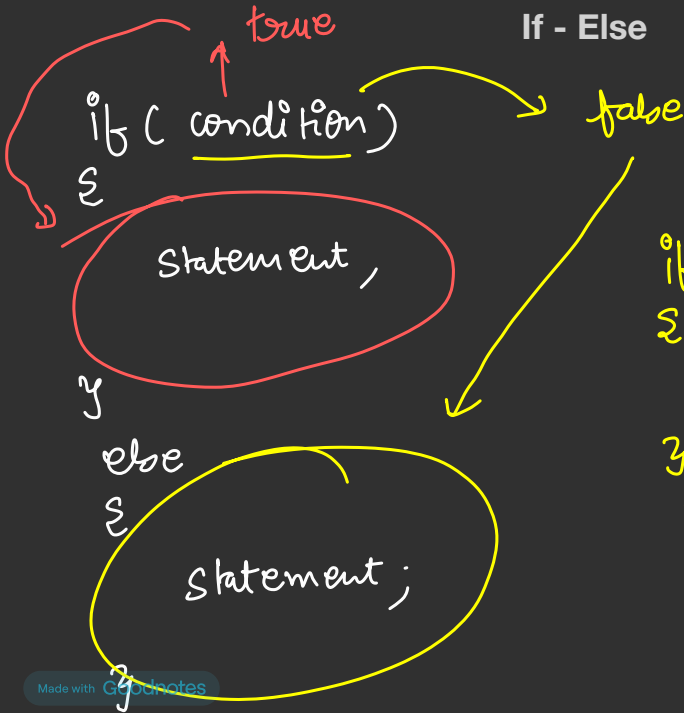
int n = 5;

if (condition)
{
 Statement
}

if (n == 5)
{
 true
}

S.O. Pln ("Number is
five");
}

If - Else



int marks = 35;

int pass = 40;

```
if ( marks >= pass )
```

```
{
```

```
    S.O Pln ("Pass");
```

```
}
```

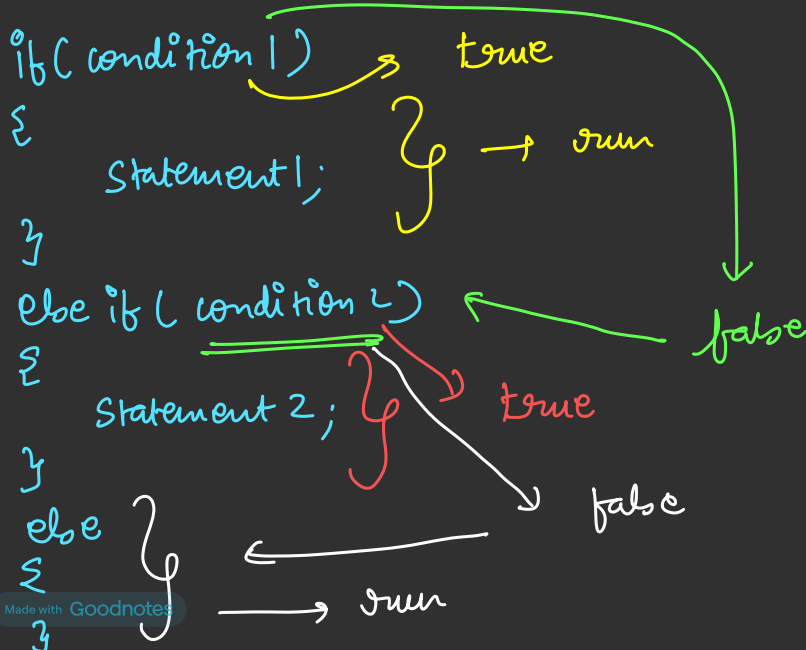
```
else
```

```
{
```

```
    S.O Pln ("Fail");
```

```
}
```

If - else - if Ladder



1 → Mon
2 → Tue
3 → Wed
4 → Thurs
5 → Fri
6 → Sat
7 → Sun

```

int a = 5; int b = 6;
if ( a > b )
{
    S.O.Pln ("Max is " + a);
}
else if ( b > a )
{
    S.O.Pln ("Max is " + b);
}
else
{
    S.O.Pln ("Equal");
}

```

$a = 5$
 $b = 5$

$a > b$
 $\rightarrow a$

$b > a$
 $\rightarrow b$

$a == b$
 $\rightarrow \text{Equal}$

Logical Operators

& & , || ,
↓ ↘
And Operator

c1	c2	<u>c1 & c2</u>
T	T	T
T	F	F
F	T	F
F	F	F

All conditions
should be true

Combination of
condition }

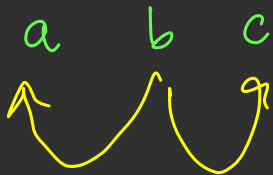
$||$ $\rightarrow$ OR operator (Any one condition will be true)

c1	c2	$c1 c2$
T	T	T
T	F	T
F	T	T
F	F	F

$!$ (condition) $\rightarrow$ true
 $\downarrow$
false
Reverse

Max of 3 numbers

a b c



a ✗

b ✗

c ✓

$$b \geq a$$

$$b \geq c$$



$$\underline{\text{max} = b}$$

$$\begin{aligned} a &= 5 \\ b &= 8 \\ c &= 4 \end{aligned}$$



$$\underline{\underline{b}}$$

```
if (a >= b && a >= c)
```

```
{  
    S.O.println(a);
```

```
}
```

```
else if (b >= a && b >= c)
```

```
{  
    S.O.println(b);
```

```
}
```

```
else
```

```
{  
    S.O.println(c);
```

```
}
```

a = 5,
b = 8,
c = 4

Ternary Operator

$a > b$? 01 : 02 ;



int maximum = $(6 > 5) ? 6 : 5 ;$



Watermelon problem Codeforces

} → HW

<https://codeforces.com/problemset/problem/4/A>

Nested if

```
if (condition 1)
{
    if (condition 2)
    {
    }
}
```

What is Switch Case ?

Keyword

↓
switch (expression)
{

case value1:

Statement;

break;

Example

default:

Statement

Stop the case

Day of week using Switch Case

Homework Questions

1. Program to check whether a number is negative, positive or zero.
2. Program to check whether a number is divisible by 5 and 11 or not.
3. Program to check whether a number is even or odd.
4. Program to find sum of all even numbers between 1 to n.
5. Program to find sum of all odd numbers between 1 to n.
6. Program to find HCF (GCD) of two numbers.
7. Program to find LCM of two numbers.

Thank You!

Please practice more questions and examples as above !!